RUI HENG YANG

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Researcher in embodied AI working on world action models and vision-language-action models for robotic manipulation, with a focus on real-time world action models. Previously developed training-free guidance methods for diffusion policies for collision avoidance in cluttered real-world environments. Research published at ICRA, IROS, ICML, and NeurIPS.

WORK EXPERIENCE

Researcher — Noah's Ark Laboratory January 2024 – Present

Projects: *Embodied AI (EAI)*

- First-authored a diffusion-based, training-free motion planner that refines trajectory priors via observation guidance at inference time, producing a **multimodal sampling distribution** for context-aware, collision-avoidant action generation. Achieved up to **56% improvement in collision-free task completion** in simulation and **80% in real-world deployments** over state-of-the-art methods in cluttered environments, using only point-to-point trajectories for training. **ICRA 2026. Patent filed.**
- Co-first-authored a closed-loop diffusion-based motion planner for robotic manipulation, enabling training-free collision avoidance via high-frequency replanning. Achieved a **5x improvement in replanning frequency** over state-of-the-art methods. **IROS 2025. Patent filed.**
- Co-developed **Ergodic Generative Flows**, a novel family of generative flow networks that leverage ergodicity to construct flows via globally defined transformations, evaluated on custom 2D imitation learning benchmarks. **ICML 2025.**
- Developed a training pipeline for diffusion-based robot controllers incorporating weak-to-strong data auto-improvement, iteratively augmenting and curating higher-quality training data using lower-capacity model outputs, across diverse simulation environments including *MetaWorld*, *Push-T*, *RoboMimic*, and *ManiSkill2*. Achieved up to **9% performance improvement** over standard training baselines.
- Built end-to-end robot control pipelines integrating perception (e.g., *YOLO-World*) and grasp generation modules (e.g., *GraspNet*) for accurate object localization and grasp pose estimation, validated through real-time human-robot interaction demonstrations.

Research Intern — Samsung AI Center (SAIC) May 2023 – December 2023

Supervisors: *Dr. Jimmy Li, Dr. Francois Hogan, Prof. Florian Shkurti, Prof. Gregory Dudek*

- Developed a generalized image compositing pipeline leveraging *DINO-ViT* for keypoint selection and feature correspondence, enabling instruction-guided background replacement via Stable Diffusion inpainting.
- Designed a tactile data collection pipeline integrating a Franka Emika arm with an RFT60-HA01 force-torque sensor and Finger-STS tactile sensor, and developed a novel force estimation algorithm combining inverse finite element methods with neural networks for precise, data-efficient force estimation.

Research Intern — Noah's Ark Laboratory May 2020 – August 2022

Supervisor*: Prof. Vahid Partovi Nia*

- Implemented benchmarking pipelines and experimental evaluations for a low-bit quantized network across image classification, object detection, semantic segmentation, and end-to-end spoken language understanding, achieving state-of-the-art performance with a **1.6x speed-up** over previous quantized SOTA networks. **ICCV 2023. Interspeech 2022.**

SELECTED PUBLICATIONS

1. Liheng Ma*, **Rui Heng Yang***, et al. Faster-WAM: Do World Action Models Need Deep Action Modules? arXiv 2026.
2. Zhiyuan Li, **Rui Heng Yang**, et al. Benchmarking Robotic Manipulation under Physically Realistic Dynamics. NeurIPS 2026 Workshop (RoboWM).
3. Amir Rasouli, Yangzheng Wu, Zhiyuan Li, **Rui Heng Yang**, et al. How VLAs (Really) Work In Open-World Environments. IROS 2026.
4. **Rui Heng Yang***, Xuan Zhao*, et al. CAPE: Context-Aware Diffusion Policy Via Proximal Mode Expansion for Collision Avoidance. ICRA 2026.
5. Sajjad Pakdamansavoji, Mozhgan Pourkeshavarz, **Rui Heng Yang**, et al. Improving Robotic Manipulation Robustness via NICE Scene Surgery. ICRA 2026.
6. Mateo Clemente*, Leo Brunswic*, **Rui Heng Yang**, et al. Two-Steps Diffusion Policy for Robotic Manipulation via Genetic Denoising. NeurIPS 2025.
7. Xi Ye*, **Rui Heng Yang***, et al. RA-DP: Rapid Adaptive Diffusion Policy for Training-Free High-frequency Robotics Replanning. IROS 2025.
8. Leo Maxime Brunswic, Mateo Clemente, **Rui Heng Yang**, et al. Ergodic Generative Flows. ICML 2025.
9. Xinlin Li, Bang Liu, **Rui Heng Yang**, et al. DenseShift: Towards Accurate and Efficient Low-Bit Power-of-Two Quantization. ICCV 2023.
10. Anderson R. Avila, Khalil Bibi, **Rui Heng Yang**, et al. Low-bit Shift Network for End-to-End Spoken Language Understanding. Interspeech 2022.

* denotes equal contribution

PATENTS

1. **Rui Heng Yang**, Leo Maxime Brunswic, Mark Chiu, et al. Methods and System for Effective Guidance Application in Fast-Generative Methods. Filed, 2026.
2. Yuhan Che, Yajun Gao, Xuan Zhao, **Rui Heng Yang**, et al. Dynamic Obstacle Avoidance Method for Closed-loop Embodied Manipulation. Filed, 2026.
3. Leo Maxime Brunswic, Mateo Clemente, **Rui Heng Yang**, et al. Efficient and Modular Flow-Matching Policy via Deconditioning. Filed, 2025.
4. Xuan Zhao, **Rui Heng Yang**, Sajjad Pakdamansavoji, et al. Predictive Failure Detection for Guided Diffusion Policy Adaptation to Uncertain Environments. Filed, 2025.
5. Xuan Zhao, **Rui Heng Yang**, Leo Maxime Brunswic, et al. Body-Aware Collision-free Motion Planning for Object Manipulation. Filed, 2025.
6. **Rui Heng Yang**, Amir Rasouli, and Yinchuan Li. Methods and System for Real-Time Guided Diffusion Policy. Filed, 2025.

EDUCATION

Master of Science in Applied Computing — University of Toronto Toronto, ON
*Artificial Intelligence - **GPA**: 4.0/4.0* *December 2023*

- **Lead Teaching Assistant for CSC110 & CSC111:** Foundations of Computer Science

B.Sc Honours Computer Science — McGill University Montreal, QC
*First Class Honours, Internship Program - **GPA**: 3.86/4.0* *Graduated May 2022*

SKILLS

Technical Skills	Python, PyTorch, NumPy, OpenCV, ROS/ROS2, Docker, Git
Languages	English (Native), French (Native), Mandarin (Native)